

First Terminal Examination - 2079

Class : XII

Subject: Mathematics (0081)

Full Marks: 75

Time: 3 hrs.

SET 'A'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternatives.

[1 × 11 = 11]

- $\frac{P(n, r)}{C(n, r)} =$
 - 1
 - r
 - r!
 - $\frac{1}{r!}$
- The number of three digits numbers in which every digit is 0 or 5 is

 - 4
 - 8
 - 9
 - 720
- In the expansion of $(1 + x)^{50}$, the sum of coefficients of odd power of x is

 - 0
 - 2^{49}
 - 2^{50}
 - 2^{51}
- The expansion $\sqrt{2 - 3x}$ is valid if

 - $x \in \mathbb{N}$
 - $|3x| < 1$
 - $|x| < \frac{3}{2}$
 - $|x| < \frac{2}{3}$
- The distance between directrices of a horizontal ellipse is

 - $\frac{a}{e}$
 - $2ae$
 - $\frac{2a}{e}$
 - ae
- The eccentricity of hyperbola $x^2 - y^2 = c^2$ is

 - 1
 - 2
 - $\sqrt{2}$
 - $\frac{1}{\sqrt{2}}$

7. Direction cosines of a line parallel to x-axis are
 - a) $k, 0, 0$ ($k \in \mathbb{R}$)
 - b) $1, 0, 0$
 - c) $0, 1, 0$
 - d) $0, 0, 1$
8. The value of 'a' for which the system of equations $x + ay = 0$, $y + az = 0$ and $z + ax = 0$ has infinitely many solutions is
 - a) 1
 - b) -1
 - c) 0
 - d) no value
9. In solving a linear programming problem, simple method is better than the graphical method if the LP problem has
 - a) no decision variable
 - b) one decision variable
 - c) two decision variables
 - d) more than two decision variables
10. In solving the system of linear equations using elementary row operations which row operations is not allowed
 - a) all entries of a row is multiplied by a non-zero number.
 - b) each element of a row is added by a multiple of another row
 - c) two rows are interchanged
 - d) each element of a row is increased by a constant number
11. Every continuous function is
 - a) increasing
 - b) decreasing
 - c) differentiable
 - d) not differentiable

Group 'B'

Short Answer Questions.

[5 × 8 = 40]

12. a) In how many ways can 4 XI students and 4 XII students be arranged at a round table if they must alternate? [2]
- b) A committee of 5 members is to be formed from 6 boys and 4 girls. In how many ways can it be done so as to include at most 3 girls? [3]

13. a) Find the term free from x in the expansion of $\left(x + \frac{1}{x}\right)^{2n}$. [2]
- b) Find the middle term or terms in the expansion of $\left(\frac{x}{a} - \frac{a}{x}\right)^{11}$. [3]
14. If $(1 + x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$, Prove that:
- i) $C_1 + C_2 + C_3 + \dots + C_n = 2^n - 1$ [1]
- ii) $C_0C_n + C_1C_{n-1} + C_2C_{n-2} + \dots + C_nC_0 = \frac{(2n)!}{n!n!}$ [4]
15. a) Find the vertices, foci and length of latus rectum of the ellipse
 $9x^2 + 16y^2 = 44$ [2]
- b) Find the equation of the hyperbola with $e = \frac{4}{3}$ and vertices $(0, \pm 7)$. [3]
16. a) If α, β & γ be the angles made by a line with the co-ordinate axes, prove that: $\cos 2\alpha + \cos 2\beta + \cos 2\gamma + 1 = 0$. [2]
- b) The direction cosines of two lines are given by $l + m + n = 0$ and $2lm + 2nl - mn = 0$. Find the direction cosines of one of the lines. [3]
17. a) Solve using Cramer's rule:
 $\frac{4}{x} + \frac{6}{y} = 0$ and $\frac{3}{x} - \frac{4}{y} = \frac{-17}{6}$ [2]
- b) Solve using row-equivalent matrix method:
 $x + z = 1$
 $2x + z = 2$
 $-5x + 9y = 3$ [3]
18. Find from first principle, the derivative of $\cos(\ln x)$. [5]
19. a) Differentiate $\cos^{-1} \frac{1-x^2}{1+x^2}$ with respect to x . [2]
- b) If $x^y \cdot y^x = 1$, prove that: $\frac{dy}{dx} = \frac{-y(y+x \ln y)}{x(y \ln x - x)}$ [3]

Group 'C'

Long Answer Questions.

[8 × 3 = 24]

20. a) In how many ways can the letters of the word "MONDAY" be arranged in the following conditions?
- i) If there is no restriction.
 - ii) If vowels are always together.
 - iii) If vowels never come together.
 - iv) If vowels can occupy only odd places. [4]
- b) Show that: $1 - \frac{1}{4} + \frac{1.3}{4.8} - \frac{1.3.5}{4.8.12} + \dots$ to $\infty = \sqrt{\frac{2}{3}}$ [4]
21. a) Find the equation of an ellipse whose one vertex is at (8, 0) and passes through (5, 6). [3]
- b) Find the direction cosines of a line joining points A(4, 3, -5) and B(-2, 1, -8). Also, show that the sum of their squares is 1. [3]
- c) If A(3, 4, 5), B(4, 6, 3), C(-1, 2, 4) and D(1, 0, 5) are any four points, find the projection of CD on AB. [2]
22. a) Mention the two steps that are used in Gauss Elimination Method to solve system of linear equations. Use it to solve.
- $$\begin{aligned}x + y - z &= -3 \\2x + y + 2z &= 16 \\x + 2y + 3z &= 15\end{aligned}$$
- [4]
- b) Using simplex method, find the optimal solution of
- $$\begin{aligned}Z &= 8x + 5y \text{ subject to} \\x + 2y &\leq 6 \\4x + 3y &\leq 12 \\x, y &\geq 0\end{aligned}$$
- [4]

❧ - The End - ❧